

1(a)	<i>pigment 1: carotene ;</i> <i>pigment 2: chlorophyll a ;</i>	2	
1(b)(i)	any eight from: <i>collect leaves from stinging nettles growing in direct sunlight and in the shade:</i> 1 <u>method</u> to select <u>plants</u> growing in direct sunlight and in the shade, using, random / systematic, sampling or <u>method</u> to grow <u>plants</u> in direct sunlight and in the shade ; 2 collect / cut / AW, <u>leaves</u> (from plants growing) in direct sunlight and in the shade ;	8	allow standardised / controlled / fixed / constant as AW for same ignore known allow: direct sunlight = high light intensity shade = low light intensity ignore no light / dark as AW for shade 1 should be 2 or more different plants allow sample plants at, coordinates / grid / quadrats, chosen using a random number generator allow sample plants at a fixed / set distance along a, line / transect R throwing quadrats ignore quadrants allow lab-based method to grow plants in different light intensities (e.g. lamp at different distances) ignore ref to controlled variables ignore ref to time (for growth) but R if less than 24 hours 2 allow if leaves are from (different parts) of same plant (in different light intensities)

Question	Answer	Marks	Guidance
1(b)(i)	3 select leaves from, same / stated, position / part / AW, of plant ; <i>use chromatography to separate the chloroplast pigments present in the leaves:</i> 4 <u>apparatus</u> to grind leaves (to prepare leaf extract) 5 use <u>pencil</u> (and ruler) to draw, origin / line / AW (on paper / TLC plate) ; 6 <u>method</u> to add a <u>concentrated</u> spot of leaf extract (to paper / TLC plate) ; 7 use, same / stated, solvent (for chromatography) or use, same type of (chromatography) paper / TLC plate ; 8 add (chromatography) paper / (TLC) plate (with spot of leaf extract), to a beaker / AW, of solvent and ensure solvent is below the, origin / line / AW ;		3 A apical leaf / same height from ground ignore same species / type / variety ignore size / number / age / colour, of leaves 4 A use pestle and mortar / blender / suitable description ignore cutting (leaves) ignore (use of) solvent / water ignore filtering / centrifuge A roll coin across leaf (onto paper / TLC plate) 5 R pen / ink <i>needs to be a method</i> 6 A use, capillary tube / melting point tube / pipette / glass rod / AW, to add leaf extract and repeat / reapply (to same position) 7 ignore solvent used for extraction allow any named organic solvent R water and ecf ignore concentration / volume 8 A from diagram allow chromatogram as AW for paper / TLC plate

Question	Answer	Marks	Guidance
1(b)(i)	9 remove, chromatogram / AW, from beaker / solvent, when solvent front is close to top (of paper / TLC plate) or after, set / stated, time ; 10 (use pencil to) mark position of solvent front (before the solvent evaporates) ; <i>collect the data needed to identify each chloroplast pigment:</i> 11 use a, ruler / AW, to, measure / note / record / AW, distance (from origin) to solvent front and distance (from origin) to, (each) spot / pigment / AW ; 12 <i>idea of</i> measure to the, centre / AW, of (each) spot of pigment ; 13 note / record / AW, colour of <u>each</u> spot of pigment;		9 allow paper / (TLC) plate as AW for chromatogram allow solvent / front as AW for solvent front ignore time if stated 10 A take photo of chromatogram 11 ignore calculate R_f / R_f formula unqualified ignore <i>ref. to repeats</i>

Question	Answer	Marks	Guidance																		
1(b)(ii)	hazard and risk and precaution ; <table><tr><th>hazard</th><th>risk</th><th>precaution</th></tr><tr><td>environment / woodland / gardens / farmland / AW</td><td>getting lost / trip hazard / injury</td><td>(wear protective) shoes / clothes / PPE stay, in a group / with expert / ranger use GPS</td></tr><tr><td>animals</td><td>allergy / irritation / injury</td><td>(wear protective) shoes / clothing / PPE stay, with expert / ranger</td></tr><tr><td>pollen / spores</td><td>allergy / irritation</td><td>mask / goggles / PPE take, anti-histamines / hayfever medication</td></tr><tr><td>knife / scalpel / AW</td><td>cuts / injury (skin / hand / AW)</td><td>cut away from the hand or cut onto a, board / tile / AW</td></tr><tr><td>(named) solvent</td><td>flammable <i>fumes / vapour:</i> irritant / toxic damage to, lungs / AW</td><td>no naked flames use fume cupboard / use well-ventilated room / keep containers covered / safety glasses / goggles / mask / PPE</td></tr></table>	hazard	risk	precaution	environment / woodland / gardens / farmland / AW	getting lost / trip hazard / injury	(wear protective) shoes / clothes / PPE stay, in a group / with expert / ranger use GPS	animals	allergy / irritation / injury	(wear protective) shoes / clothing / PPE stay, with expert / ranger	pollen / spores	allergy / irritation	mask / goggles / PPE take, anti-histamines / hayfever medication	knife / scalpel / AW	cuts / injury (skin / hand / AW)	cut away from the hand or cut onto a, board / tile / AW	(named) solvent	flammable <i>fumes / vapour:</i> irritant / toxic damage to, lungs / AW	no naked flames use fume cupboard / use well-ventilated room / keep containers covered / safety glasses / goggles / mask / PPE	1	ignore low / medium / high, risk R no risk ignore harmful unqualified ignore plants / stinging nettles / trichomes as hazard ignore wearing gloves as a precaution allow check yourself / immunise (as precaution against ticks) allow epipen as precaution for allergies NOTE: allow away from body ignore cuts clothing / AW ignore cutting leaves / plant unqualified for hazard ignore water as hazard ignore allergy / corrosive as risk for solvent
hazard	risk	precaution																			
environment / woodland / gardens / farmland / AW	getting lost / trip hazard / injury	(wear protective) shoes / clothes / PPE stay, in a group / with expert / ranger use GPS																			
animals	allergy / irritation / injury	(wear protective) shoes / clothing / PPE stay, with expert / ranger																			
pollen / spores	allergy / irritation	mask / goggles / PPE take, anti-histamines / hayfever medication																			
knife / scalpel / AW	cuts / injury (skin / hand / AW)	cut away from the hand or cut onto a, board / tile / AW																			
(named) solvent	flammable <i>fumes / vapour:</i> irritant / toxic damage to, lungs / AW	no naked flames use fume cupboard / use well-ventilated room / keep containers covered / safety glasses / goggles / mask / PPE																			